

Barrier Distribution in ${}^6\text{Li} + {}^{209}\text{Bi}$ Fusion Reactions

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December 10, 2024

Abstract

Barrier distributions are crucial for understanding the dynamics of nuclear fusion reactions. This paper investigates the barrier distribution for the ${}^6\text{Li} + {}^{209}\text{Bi}$ system. Using precise experimental measurements, the barrier distribution was analyzed to understand the effects of channel coupling, projectile breakup, and sub-barrier tunneling on the fusion process. These insights provide a deeper understanding of weakly bound projectile interactions and their impact on nuclear fusion.

1 Introduction

The study of nuclear fusion, especially involving light projectiles such as ${}^6\text{Li}$, plays a crucial role in understanding nuclear reactions, astrophysical processes, and energy generation mechanisms. One of the key tools in studying these fusion reactions is the analysis of barrier distributions. Barrier distributions provide detailed insights into the fusion dynamics, revealing the effects of nuclear structure, quantum tunneling, and coupling between reaction channels.

In the case of ${}^6\text{Li}$ and ${}^{209}\text{Bi}$, the weakly bound nature of ${}^6\text{Li}$ adds an extra layer of complexity, introducing additional phenomena such as breakup and coupling to multiple reaction channels. This paper focuses on analyzing the barrier distribution derived from experimental data for this reaction, highlighting its significance in understanding sub-barrier fusion and the associated dynamics.

2 Barrier Distribution: Theoretical Context

Fusion reactions are governed by the interaction potential between colliding nuclei, comprising:

- **Coulomb repulsion:** Due to the positively charged nuclei.
- **Nuclear attraction:** From short-range nuclear forces.
- **Centrifugal barrier:** Arising from angular momentum.

For a given system, these interactions produce a potential barrier that must be overcome for fusion to occur. Barrier distributions, obtained as:

$$D(E) = -\frac{d^2(E\sigma)}{dE^2},$$

provide a detailed representation of the energy-dependent probability of overcoming these barriers.

The presence of multiple barrier heights in the distribution is a direct consequence of:

- **Channel Coupling:** Vibrational, rotational, or breakup states of the interacting nuclei contribute to the distribution, resulting in multiple peaks.
- **Quantum Tunneling:** At sub-barrier energies, the fusion probability is non-zero due to tunneling, which is sensitive to the shape and height of the potential barrier.
- **Projectile Structure:** The weak binding energy of ${}^6\text{Li}$ introduces breakup channels, which influence the distribution's shape.

3 Experimental Methodology

The barrier distribution for the ${}^6\text{Li} + {}^{209}\text{Bi}$ system was determined through precise measurements of the fusion excitation function. The experiment involved a ${}^6\text{Li}$ beam directed at a ${}^{209}\text{Bi}$ target, with fusion cross-sections measured across a wide range of center-of-mass energies (E_{cm}). Gamma-ray spectroscopy of fusion residues was used to identify successful fusion events.

The fusion excitation function was analyzed, and the barrier distribution was extracted using:

$$D(E) = -\frac{d^2(E\sigma)}{dE^2},$$

where $D(E)$ represents the barrier distribution. This method provides a smoothed representation of barrier probabilities across energy ranges.

4 Results and Discussion

The barrier distribution graph for the ${}^6\text{Li} + {}^{209}\text{Bi}$ system reveals several key features:

- **Multiple Peaks:** The distribution shows distinct peaks corresponding to the coupling of nuclear, vibrational, and rotational states. These peaks indicate different fusion pathways, each associated with a unique barrier height.
- **Sub-barrier Fusion:** Significant fusion probabilities were observed at energies below the average Coulomb barrier, highlighting the role of quantum tunneling in this weakly bound system.
- **Broadening Effects:** The barrier distribution is notably broader than those observed for tightly bound projectiles. This broadening reflects the influence of projectile breakup and coupling effects, which reduce the effective barrier and increase fusion probabilities at sub-barrier energies.

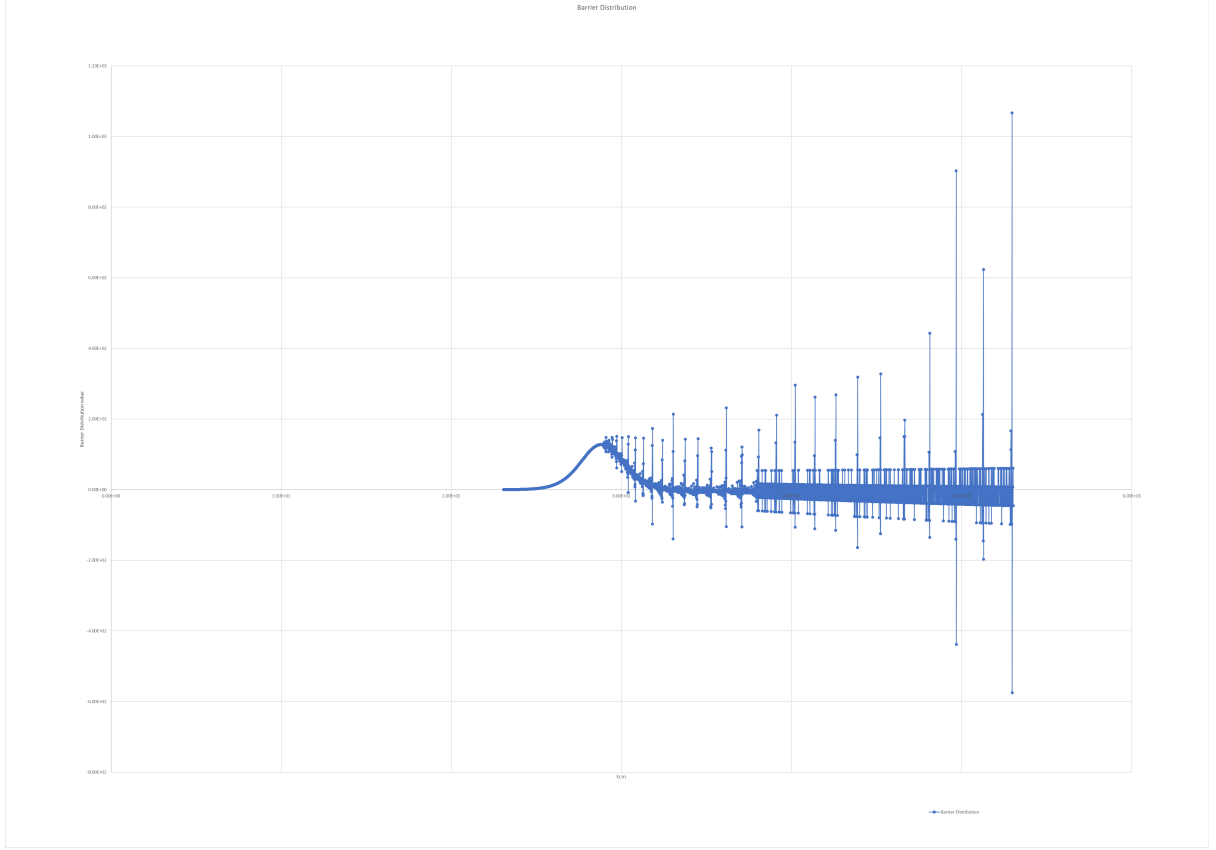


Figure 1: Barrier distribution for ${}^6\text{Li} + {}^{209}\text{Bi}$. The graph shows the energy-dependent probability of overcoming the fusion barrier.

The results emphasize the importance of projectile breakup, a characteristic of ${}^6\text{Li}$, which leads to channel coupling and energy dissipation. These effects make the barrier distribution for this system unique compared to reactions involving more tightly bound nuclei.

Additionally, the following formula was used in the analysis to compute the barrier distribution, which accounts for various factors such as energy, cross-section, and channel coupling:

$$D(E) = \frac{(D_6 \cdot E_6) - (E_{\text{cm}} \cdot 2 \cdot \sigma_{\text{fusion}}) + (D_8 \cdot E_8)}{0.001},$$

where:

- D_6 and E_6 are terms related to energy and cross-section from experimental data.
- E_{cm} represents the center-of-mass energy.
- σ_{fusion} is the fusion cross-section.
- D_8 and E_8 represent additional energy and coupling factors.

4.1 Application to Barrier Distribution

Barrier distributions are often calculated using the second derivative of the fusion cross-section with respect to energy, represented as:

$$D(E) = -\frac{d^2(E\sigma)}{dE^2}.$$

This formula approximates the calculation of barrier distributions by combining experimental values and adjusting for scaling. It uses numerical differentiation techniques to obtain a smooth representation of the energy-dependent barrier probability. In this context, the formula above accounts for experimental parameters such as energy and fusion cross-section, incorporating them into a numerical method for calculating the barrier distribution. This approach is crucial in understanding how nuclear structure, quantum tunneling, and channel coupling affect the fusion reaction at various energies.

5 Conclusion

The analysis of barrier distributions for the ${}^6\text{Li} + {}^{209}\text{Bi}$ fusion reaction offers a comprehensive understanding of the interplay between quantum tunneling, nuclear structure, and channel coupling. The presence of multiple peaks and sub-barrier fusion probabilities underscores the impact of projectile breakup on fusion dynamics. Future work focusing on advanced coupled-channel calculations and high-resolution experimental data can provide deeper insights into the mechanisms driving such reactions.

This investigation enhances our understanding of fusion reactions involving weakly bound systems, with implications for nuclear physics and astrophysical studies.